



Flying Liferaft

Simulation and Testing of UAV for Emergency Aid Deployment at Sea



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Group Design Project

H401 MEng Aeronautical Engineering

20/06/2022

FLR22

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Abstract

This report focuses on the final stages of the design life cycle, specifically on simulation and testing of the design through the use of X-Plane 10 and the Imperial College Flight Simulators. In this report, the testing methods and the different results extracted from each test are detailed, including also the modelling of the aircraft in the Plane-Maker tool. The performance tests performed were steady level flight test to determine the lift curve slope, drag polar, cruise performance, endurance performance and also climb performance of the aircraft. Stall tests were also performed, determining the stall AoA of 15.1° and a stall speed of 23.3 m/s. The lift curve slope, stall AoA and $C_{L_{MAX}}$ also matched with relatively small percentage errors with predictions, while other performance metrics had been swayed by the modelling of the aircraft, namely the drag characteristic definition of the aircraft. In addition, to these tests, since the UAV has been designed in a hybrid configuration (in terms of propulsion, but also VTOL fixed-wing layout) the takeoff and transition performance metrics were also assessed. Another iteration of the simulation testing will be performed with a more accurate drag characterisation in X-Plane in order to confirm the predicted values before presenting the design to clients.

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List of Symbols

V_e	Equivalent Airspeed
C_L	Lift Coefficient
α	Angle of Attack
S_{ref}	Wing Reference Area
$S_{frontal}$	Frontal Area
$C_{L_{MAX}}$	Maximum Lift Coefficient
$VTOL$	Vertical Takeoff and Landing
$eVTOL$	Electric Vertical Takeoff and Landing
UDP	User Datagram Protocol
K_n	Stick Fixed Static Margin
$d\epsilon/d\alpha$	Downwash derivative with respect to angle of attack
V_x	Best angle of climb velocity
V_y	Best rate of climb velocity

1 Introduction

The design brief for this project was to design and test an unmanned autonomous vehicle which will have the primary purpose of bringing aid to people stranded at sea, specifically focusing on refugees crossing the Mediterranean Sea. Accordingly for this mission, a suitable configuration was designed with sufficient performance capability and also sufficient payload capacity for flexibility in carrying different amounts and different types of payload such as buoys or a life raft. Through a conceptual design phase, a hybrid UAV with eVTOL capability was found to be most suitable for this mission.

Once the design phase had kick-started, simulation models were created and prepared for testing. The modelling methods and testing methods, as well as the results will be explained in further detail in this report.

2 X-Plane Model

X-Plane 10 was the flight simulation software used for the modelling and testing of the aircraft. In the early stages of the project, focus was put on becoming familiar with the simulation software and also the modelling software tools, Plane-Maker and Airfoil-Maker. As well as familiarisation with the tools, understanding the physics behind the simulation software was also required in order to model the aircraft as accurately as possible and also to explain any discrepancies in test results with design values.

2.1 X-Plane Physics and Limitations

X-Plane uses blade element method, rather than stability derivatives as other simulation software use. This means the aircraft geometry is broken down into elements and a force build up is performed, by evaluating the velocities and angles of attack at each element. This provides a more detailed characterisation of the aircraft than stability derivatives. [1] The use of blade element method as opposed to the lifting line theory and vortex lattice methods used by the Aerodynamics team, could also cause differences in the aerodynamic characteristics to arise.

Due to the nature of how blade element method is applied, some aerodynamic effects cannot be accurately modelled. For example, turbulent dynamics cannot be modelled. Focusing on the modelling more relevant to this aircraft, the wake dynamics and prop wash are modelled to some extent but this is using empirical data, which may not be accurate for this aircraft due to its smaller size compared to the aircraft that the developer focused on. In addition, the simulator does not model entrainment between propwash and the outer flow region, which could overestimate the performance of the wing and tail lifting surfaces. [2] Although this source is a recent video, and therefore applies more to the newer version X-Plane 11, the prop wash effects, for example, were seen to be modelled in X-Plane 10 through the physics model visualisation and also flying the aircraft with VTOL propellers turned on, which caused reduction in lift as expected. However the accuracy of this effect could not be analysed quantitatively.

2.2 Aircraft Modelling

Airfoils are defined through the Airfoil-Maker tool, which is based on curve fitting data relating to the lift curve slope, drag polar and pitching moment coefficient data for each aerofoil used. For these sets of data, Airfoil tools was used for the different airfoils required. The process of curve fitting is subject to human error, however there is a lot of detail involved in this tool for the airfoil coefficient determination. Figure 1 shows the

Airfoil-Maker tool with the plots from Airfoil tools which characterise the lift curve, drag polar and also the pitching moment coefficient curves .[3]

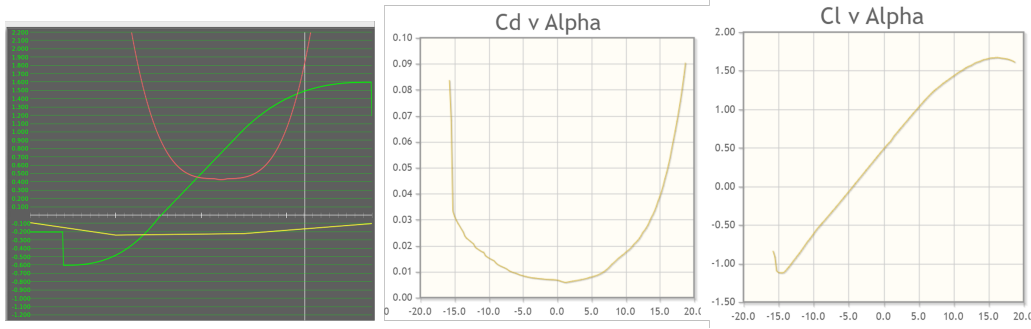


Figure 1: Airfoil-Maker Tool with plots used for curve fitting.

Plane-Maker is the main modelling software tool in this package, where the whole geometry and sub-systems of the aircraft were designed. Parts, such as the booms which connect the propellers to the wing are modelled as miscellaneous bodies. The final model of the aircraft can be seen in Figure 2.

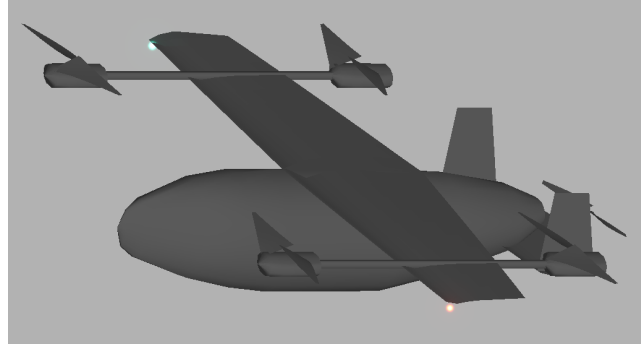


Figure 2: Final Plane-Maker Model.

As the propulsion method chosen for the aircraft was electric and petrol hybrid, this had to be modelled in Plane-Maker. On this software tool, there is not a lot of scope for battery engineering as the batteries are limited to a voltage of 24 V. In addition, the simulator does not offer capability exclusively for eVTOL, but does offer the flexibility of including electric engines which can run off the battery defined by the designer. By doing this, during simulation the battery voltage can be seen decreasing but the current is at a value that is defined in Plane-Maker. A constant current draw is unrealistic so the power that was planned to be calculated from this would be also inaccurate. Since the main goal of modelling electric engines was to analyse the energy consumption and therefore to validate the size of the battery, for the testing of the eVTOL engines during takeoff and transition, the engine power output from X-Plane was used and integrated with respect to time to calculate energy consumption. Additionally, a lower transmission loss of 0.2 was applied to model the electric engines as they are more efficient (up to 85%) than piston engines. [4]

3 Simulink to X-Plane Interface

In order to interface with X-Plane and collect data to analyse, the Simulink software tool was used which allowed communication with X-Plane over internet through the UDP

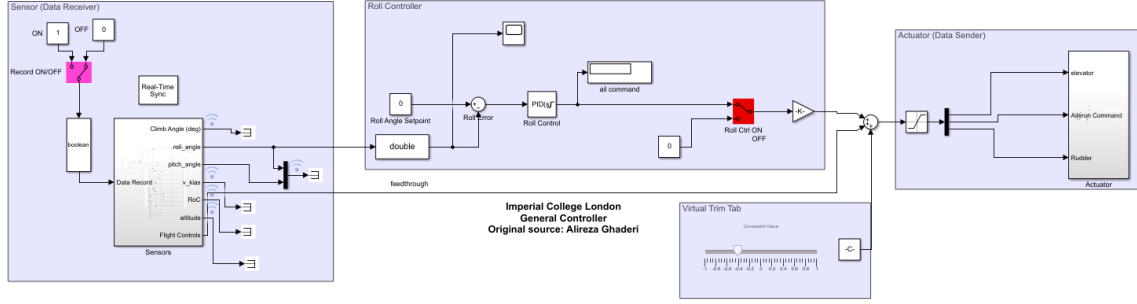


Figure 3: Simulink Testing Model for Performance and Stability Tests (in fixed wing mode).

communications protocol.

3.1 Flight Testing Simulink Model

For the majority of the flight tests performed, the aircraft model in X-Plane was modelled as a purely fixed-wing aircraft with the VTOL propellers represented as folded into a single body. Therefore one throttle command was used, along with the flight control commands. The commanding was able to be done through simply through X-Plane alone without sending commands from Simulink, however due to how the the aircraft was modelled with only the pusher propeller, there was a gyroscopic effect seen during flight which created the tendency of the aircraft to roll to one side. In order to mitigate this effect, a simple PID roll controller was added to the Simulink model to keep a zero roll angle. Similarly, for trimming the aircraft a virtual slider for pitch control was added in Simulink. These features within the whole Simulink model can be seen in Figure 3.

3.2 VTOL Throttle Command

One issue faced when modelling the full aircraft was the need for 5 throttle inputs into X-Plane. This was not an issue on its own, as in the real world, the pilot on standby will not be controlling all 5 throttles individually, but rather sending throttle and flight control commands which will be evaluated by a motor mixing algorithm. One way to accomplish this was using the X-MultiThrottle plugin, which could bind multiple engines to a single throttle command. [5] However, this tool had limited flexibility and the code had to be modified to facilitate for this aircraft.

Since Simulink was used for flight testing, the same tool was decided to be used for the control (as well as testing) of the VTOL phase too, instead of the plugin. The Simulink model used for the VTOL control can be seen in Figure 4, which shows a high level configuration of the system with colour coding of subsystem groups: blue for sensors, purple for the controller from the controls team, red for the manual override control with a joystick and green for the actuator subsystem.

4 Flight Testing Program

The flight test program was split into different parts, focusing on performance, stability and also the autonomous capability of the system. These tests were carried out at every major design decision in the project timeline in the form of verbal feedback, however the majority of the numerical results were processed as the design approached final stages.

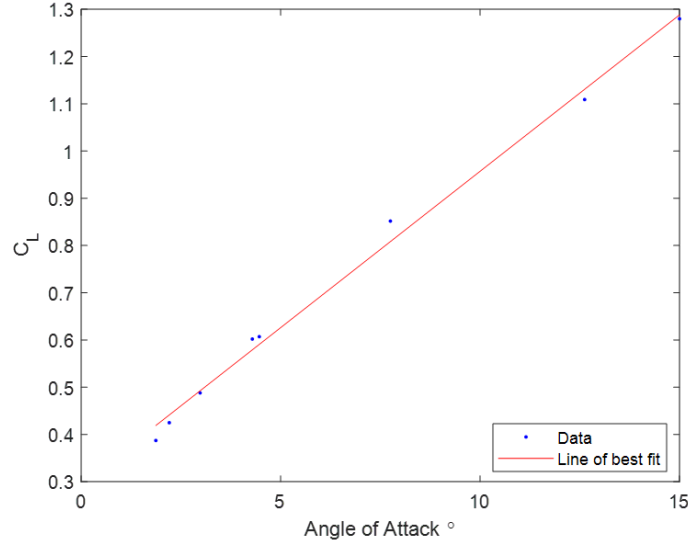


Figure 5: Lift Curve from Flight Simulation.

aircraft during stall, which means the lift would not be exactly equal to the weight of the aircraft, and potentially lift would be greater than the weight during stall if the aircraft is falling in the air causing a greater angle of attack, resulting in the lower stall speed seen. This could explain the lower stall speed from flight testing and slightly higher stall α . In addition, X-Plane does not model vortices forming in the wake during separation of flow, and this could also result in later stall being noticed.

4.2 Steady Level Flight

This test was used to define the lift curve slope of the aircraft, by testing the aircraft at different speeds to cover its range of speeds. As these tests were performed at trim conditions and level flight, the weight of the aircraft can be used to calculate the lift coefficient, C_L , at each speed.

Parameters	Flight Test Data	Analysis Data	Percentage Difference
Zero-lift α (°)	-0.442	-2.71	83.7%
Lift Curve Slope (/rad)	3.79	4.71	19.5%

Table 2: Lift Curve Characteristics Comparison with analysis by Aerodynamics Team.

As seen in Table 2, the percentage differences between analysis and flight test data are relatively large, which could be due to the modelling of the wing and lifting components in Plane-Maker and Airfoil-Maker tools due to the curve-fitting method or other modelling assumptions made by Blade Element Method, which are portrayed more accurately by the CFD simulations.

4.3 Drag Calculation

At trimmed flight conditions, with the speed of the aircraft steady, the drag characteristics can also be evaluated, by measuring the thrust output, again at the different speeds through the range of speeds. This allowed the drag polar curve to be created, which can be seen in Figure 6.

There is a significant percentage difference seen in the C_{D0} and L/D_{max} , which suggests an issue with drag modelling. In order to analyse the cause of this difference in more

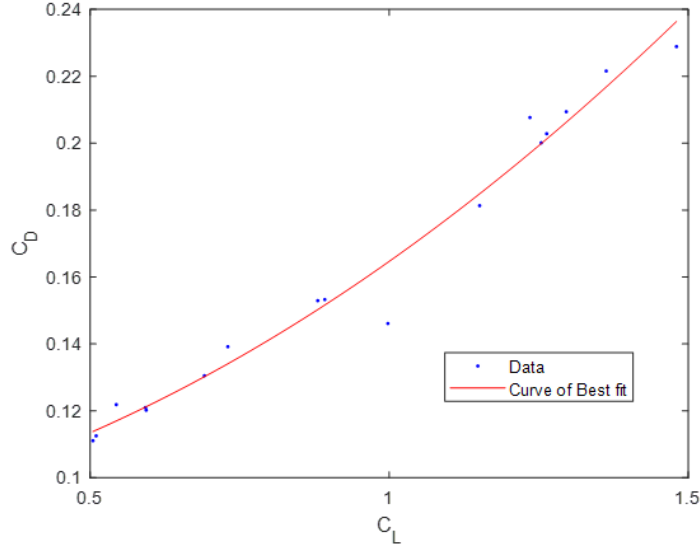


Figure 6: Drag Polar from Flight Simulation.

Parameters	Flight Test Data	Analysis Data	Percentage Difference
C_{D0}	0.0861	0.0660	50.8%
L/D_{max}	6.75	8.43	24.9%

Table 3: Drag Characteristics Comparison with analysis by Aerodynamics Team.

detail, the modelling in Plane-Maker was checked for errors. The simulator seems to be over-estimating the drag on the aircraft, which could be due to how the C_{D0} value entered in Plane-Maker is used. Through initial tests the culprit was found to be the fact that the C_{D0} value in Plane-Maker is based on frontal area, rather than the wing reference area, which is what the C_{D0} from the Aerodynamics team is referenced to. Therefore initial tests gave an unrealistic and optimistic drag value compared to the Aerodynamics team. A better way for the C_{D0} for the fuselage was chosen as 35 % of the overall aircraft C_{D0} , based on empirical data. [7] After also increasing the fuselage C_{D0} by the ratio of $S_{ref}/S_{frontal}$, the changes seen in the drag during level flight (and therefore the drag coefficient values) were significant. This suggests that the fraction of total drag accounted for by the fuselage may be too big from empirical data or some of the drag modelling limitations of X-Plane mean that the fuselage drag coefficient is used in an unexpected way by the software to account for the features not characterised by Blade Element theory.

4.4 Endurance and Cruise Performance

Endurance and cruise performance are major technical performance measures for this aircraft as the application requires flexible range and endurance metrics. For these tests, the fuel flow rate was measured at different speeds when the aircraft was trimmed. Due to difficulty experienced with trimming the aircraft, even with the virtual trim tab for different speeds, many tests were performed in order to fit a reliable curve of best fit to extract further information from.

For the best endurance speed, the speed at which the lowest fuel flow rate was required can be used. By carrying out the test at many different speeds across the speed range, a curve of best fit based on empirical observations can be fitted and the best endurance speed can be extracted at the minimum point of the curve, as seen in Figure 7.

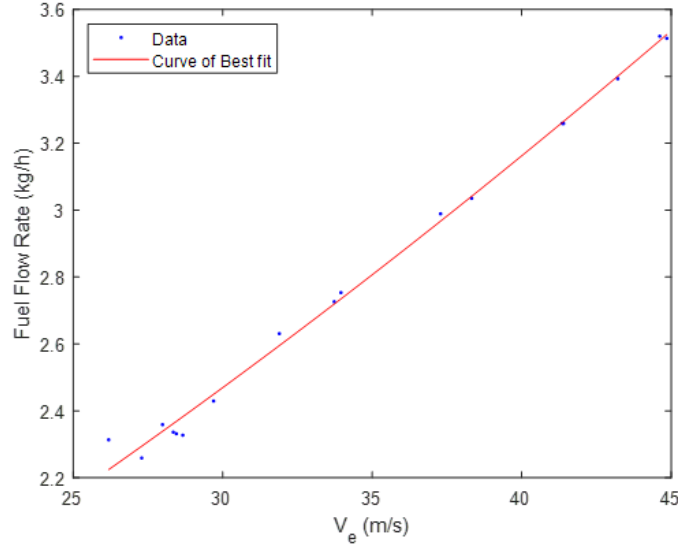


Figure 7: Endurance Performance from Flight Simulation.

Likewise, for best range speed another metric was used in order to identify the best cruise speed. This metric was the specific air range (SAR) which gives the amount of distance travelled per unit mass of fuel consumed. From the flight test results, a parabolic shape for the SAR curve is seen, however the maximum SAR value is seen at the maximum speed of the aircraft, which suggests that it is best to travel at the top speed for maximising range and also time to target. This can be explained by the the specifications of the engine type used, which has better fuel efficiency when working at its design RPM value compared to lower RPM values. [8]

The range plot shown in Figure 9, shows that a maximum range of 64.1 km can be achieved. It should be noted, however, that this range is the worst-case scenario taking into account also the maximum possible headwind speed of 12 m/s in the Mediterranean. [9]

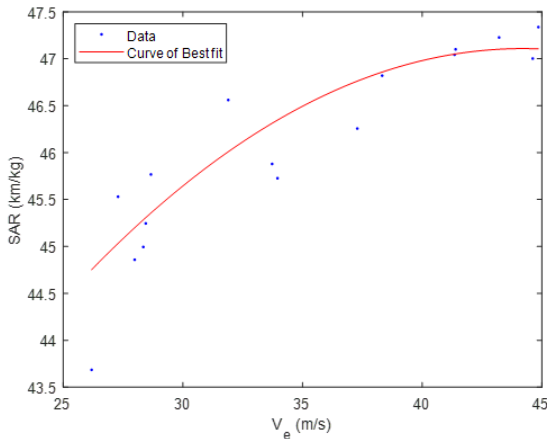


Figure 8: Specific Air Range Plot.

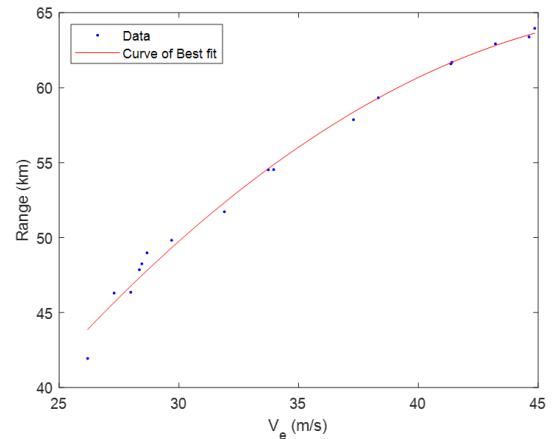


Figure 9: Range vs Speed Plot.

As well as comparing the flight test results with the targets specified from the customer requirements, the cruise and endurance performance data were also compared with relations from literature.

Parameters	Flight Test Data	Target	Percentage Difference
V_{eBR} (m/s)	44.8	58.3	23.1%
V_{eBE} (m/s)	27.3	41.8	34.9%

Table 4: Comparison with targets set out from conceptual design and initial sizing.

For piston-engine aircraft, the power available is approximately proportional to the fuel flow rate, and therefore the V_{eBE} is the minimum power speed ($0.76V_{eMD}$). Similarly the V_{eBR} is the speed which maximises the speed to power ratio and therefore is equal to the minimum drag speed V_{eMD} . [10]

By calculating the lift coefficient from minimum drag C_{LMD} , the speed for minimum drag, V_{eMD} , can also be calculated using steady level flight lift equations. Using the literature relations from [10], the minimum power speed can also be calculated. The results and comparison can be seen in Table 5.

Parameters	Flight Test Data	Literature based Values	Percentage Difference
V_{eBR} (m/s)	44.8	29.5	51.9%
V_{eBE} (m/s)	27.3	22.8	19.7%

Table 5: Comparison with targets set out from conceptual design and initial sizing.

From the comparison with literature, the values from flight testing do not seem to agree. However, it must be noted that the literature relation is based upon the fact that for piston aircraft the power available is proportional to the fuel flow rate. This was not the case in the simulation, as even with no thrust being produced there was still fuel flow occurring. This could arise to a number of reasons, such as the efficiency of the propellers in the simulations being inefficient.

In addition, the literature relations give that the best loiter speed is around 22.8 m/s, which is below our stall speed, and therefore is impractical to fly at. Therefore the literature relations were used merely as a point of reference, but in terms of the general trend of the SAR and SE plots seen above, they seem to follow the literature trend from empirical data. [10]

4.5 Climb Performance

For analysing climb performance of the aircraft, two different types of tests are possible to conduct: level acceleration and sawtooth climbs. Both of these tests are based on energy methods and specifically calculating the specific excess power of the aircraft.

Initially the level acceleration test was used. However after doing a set of tests, it was evident that sawtooth climb tests will provide a bigger dataset which can filter out any errors from testing. This is due to the level acceleration test measuring acceleration of the aircraft at different altitudes, and therefore is more suitable for high performance aircraft which have a larger speed range, especially ones that operate within both subsonic and supersonic regimes. [11] This aircraft had a relatively small speed range, and therefore the level acceleration test was very susceptible to pilot errors due to the large percentage errors that arose in the accelerations between tight speed bands. Therefore the sawtooth climb test method was adopted.

The sawtooth climb test was performed by climbing through different altitude bands at steady airspeed and calculating the climb rate, which in turn gives the specific excess

power by Equation 1, since the horizontal acceleration component was kept at zero during the steady climb. The resulting plots for specific excess power against equivalent airspeed and the specific excess power contours can be seen in Figures 10 and 11, respectively. The purpose of Figure 10, is to be able to extract the best rate of climb velocity (V_y) and also the best angle of climb velocity (V_x) for each altitude. For V_y , this can simply be done by looking at the speed at which the specific excess power is maximised for each curve. For V_x , the approach would be to draw a tangent from the origin of the graph to each of the curves. [11]

The differences between the performance analysis and the flight test data, especially in the maximum ceiling value of the aircraft, can arise due to 2 possible reasons for this test: either the drag is being modelled too pessimistically or the thrust is being modelled by X-Plane as being too low, hence hinting that the propeller geometry and efficiency may not be accurately modelled in the simulator as the team had designed. However, a common cause for mismatch between the predictions and the flight test data has been the drag and therefore that is something that has likely affected the climb performance. At the same time, a smaller deviation is seen in the climb rate, which suggests there is less specific excess power than predicted. The smaller percentage difference seen in the maximum climb rate, as opposed to the maximum ceiling shows that the specific excess power available is close to the predicted value (even though it is lower in the flight tests), however the influence of altitude on specific excess power has been modelled differently in analysis and in the simulation. This can also be due to the modelling of the propeller geometry in Plane-Maker.

$$P_s = \frac{dh}{dt} + \frac{V}{g} \frac{dV}{dt} \quad (1)$$

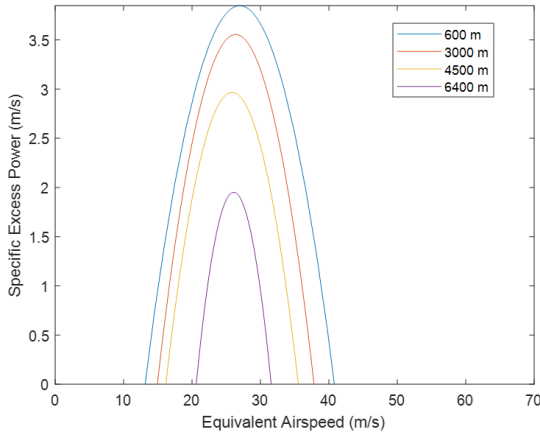


Figure 10: SEP vs Speed.

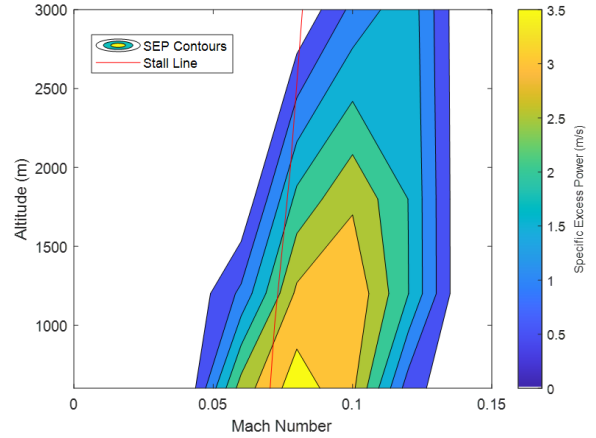


Figure 11: Specific Excess Power.

Parameters	Flight Test Data	Analysis Data	Percentage Difference
Maximum Climb Rate (m/s)	3.50	4.38	20.1%
Maximum Ceiling (m)	3600	6330	43.1%

Table 6: Climb Performance Comparison with analysis by Performance Sub-Team.

The differences between the performance analysis and the flight test data, especially in the maximum ceiling value of the aircraft, can arise due to 2 possible reasons for this test: either the drag is being modelled too pessimistically or the thrust is being modelled by X-Plane as being too low, hence hinting that the propeller geometry and efficiency

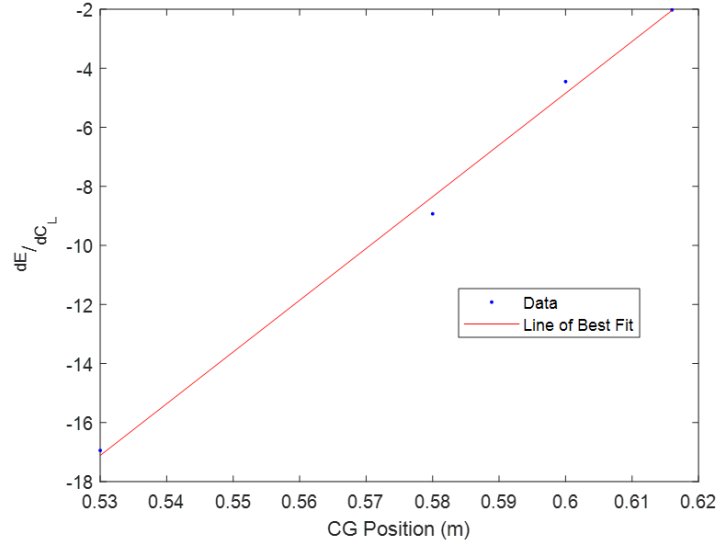


Figure 12: Elevator Deflection - Lift Coefficient slope Plot for finding Static Margin.

may not be accurately modelled in the simulator as the team had designed. However, a common cause for mismatch between the predictions and the flight test data has been the drag and therefore that is something that has likely affected the climb performance. At the same time, a smaller deviation is seen in the climb rate, which suggests there is less specific excess power than predicted. The smaller percentage difference seen in the maximum climb rate, as opposed to the maximum ceiling shows that the specific excess power available is close to the predicted value (even though it is lower in the flight tests), however the influence of altitude on specific excess power has been modelled differently in analysis and in the simulation. This can also be due to the modelling of the propeller geometry in Plane-Maker. Nevertheless, the maximum climb gradient of 3.5 m/s meets the EASA Early Access Rules for UAVs, which give a minimum climb gradient of 500 ft/min (2.54 m/s). [12]

4.6 Longitudinal Stick-Fixed Static Stability

As well as the performance tests, the longitudinal static stability was also assessed through flight simulation. The stick fixed case was focused on for static stability as the stick-free case is more relevant for manual control with pilot in the loop. As this system will be fully autonomous, except during manual override, it is more useful to determine the amount of control and elevator deflection required for trim, rather than the force required to do so. [13]

Having decided to carry out stick-fixed stability testing, data of the elevator deflection at different trim conditions was recorded. This also was done at 4 different CG positions, including the aft-most CG location of 0.616 m. The plots from the individual tests for each CG position can be seen in Figures 13 and 14 in the Appendix. From the elevator deflection - lift coefficient slope plot, as seen in Figure 12, the neutral point and static margin for the aircraft were calculated by finding the CG position at which the slope would equal zero, where there will be zero pitching moment.

The effects of a relatively low static margin were experienced while flying the aircraft on the flight simulators: the aircraft was very responsive, especially to pitch commands and easily made unstable due to this effect. This was somewhat resolved during flight testing

by adding a sensitivity gain on the Simulink model in order to adequately moderate the pilot input, and therefore also the response of the aircraft.

Parameters	Flight Test Data	Analysis Data	Percentage Difference
Neutral Point(m)	0.628	0.660	5.20%
K_n	0.0348	0.136	291%

Table 7: Static Stability Comparison with analysis by Stability Sub-Team.

There is a significantly large difference seen in the static margin between the analysis and flight test data. When this issue was discussed further, it was evident that the analysis was based upon stick-free static stability, whereas during flight testing the stick-fixed case was examined due to the type of application of this drone, as mentioned before.

4.7 VTOL Takeoff and Transition Tests

Flight tests for UAVs are not well defined as they are still new to aviation. However, from literature, 4 types of tests are usually performed for UAV systems: validation flights, documentation tests, payload directed flight tests and mission focused flight tests. [14]

The documentation and payload directed tests are more relevant when prototyping starts. At this stage of the design the validation and mission focused tests were performed. One important aspect of the UAVs mission is to be able to takeoff from a ship and safely transition in to cruise flight. This was one of the aspects of the mission profile that was tested in order to validate the controller from the Controls team. For this, the Simulink model shown earlier, in Figure 4. Key metrics to be tested during the takeoff and transition phases were the energy consumption and the settling time of the manoeuvre. The energy consumption was a key design driver throughout the project, especially for the battery sizing, and the safety of the UAV during takeoff from ship can be evaluated by the accuracy in following a set trajectory. This was measured by integrating the power usage of the 4 eVTOL engines over time in order to find the total energy. The results from the controller testing can be seen in Table 8.

Parameters	Flight Test Data	Predicted Data	Percentage Difference
Energy Consumption (J)	1.65×10^5	8.00×10^4	110%
Settling Time (s)	-	10-15	-

Table 8: Takeoff (only) Controller Performance with predictions by Controls Team.

The controller did not settle at the required altitude of 12 m and therefore a settling time cannot be quoted. However, the energy consumption up until the altitude required was calculated using the method discussed above, and there was a much larger power consumption seen from the flight test than predicted. When exploring the reason for this difference further, it was clear that the prediction was made using a look up table for power supplied by the Propulsion Team. In addition to this, the energy used during idle operation and also the spool up time modelled in X-Plane can cumulatively add to create this percentage difference, especially when the differences in engine performances are multiplied by 4 (with 4 eVTOL engines).

As the controller for the transition phase was not finalised at this stage of the design, the transition was done manually, using the same Simulink model from Figure 4, in order to test the manual override handling qualities. However, again the key quantitative

performance metric assessed here was the energy consumption over the whole transition phase, for which the comparison can be seen in Figure 9.

Parameters	Flight Test Data	Predicted Data	Percentage Difference
Energy Consumption (J)	5.51×10^5	3.92×10^5	40.6%

Table 9: Takeoff Transition Performance with predictions by Controls Performance Team.

The differences seen here can be justified by the fact that with manual control, perfect level transition as modelled by the performance is difficult to achieve. In addition, the prop-wash effects and interaction between streamlines through the different propellers can increase the energy required to transition, which were not modelled during predictions.

5 Conclusions

From the flight testing and experience flying the aircraft in the flight simulators, the aircraft is relatively difficult to handle at first but with practice the handling was manageable. However, this is an autonomous system and so the handling qualities are more relevant for manned aircraft.

In terms of performance, the flight test results show that the aircraft model was not as performant as expected, due to drag modelling issue. This is likely due to the drag coefficients entered in Plane-Maker, which may have been too high. Following this iteration, the Plane-Maker model will be re-assessed in order to produce more accurate results. The accuracy can be assessed by comparing the drag test data to that of the STAR-CCM analysis. The major metrics in disagreement are the drag influenced parameters. These values show that drag is over-estimated in the model than theory predicts. The C_{D0} for the fuselage was taken as a percentage of that total value from empirical data, which may have been too large for this type of aircraft. [7] To mitigate this, a more detailed drag breakdown can be made from STAR-CCM analysis so that the drag can be accurately modelled for the fuselage as that is the major drag causing body. The area scale factor for converting the drag coefficient from STAR-CCM to Plane Maker will also be checked again and modified if required for further testing. The STAR-CCM simulation must be taken as the reference, as it models aerodynamic characteristics to a higher level of accuracy than blade element method.

There were also some disagreements in the static stability metrics, in particular the static margin, which was analysed to be higher by an order of magnitude. As mentioned, from experience with flying the aircraft, the relatively low static margin can be justified as the aircraft was difficult to control in pitch and also very responsive. Therefore the reason for the discrepancy could be due to the different methods used (in flight test, the elevator deflection is directly used but in theoretical analysis the pitching moment coefficient is used along with efficiency factors and $d\epsilon/d\alpha$ terms), which are not necessarily taken into account in the model.

Overall, although certain modelling errors are evident to be present in the simulation, namely the C_{D0} values, the test methods applied and the two rounds of tests performed with the aircraft with slightly different design parameters, the effect of certain design decisions on flight simulation and flight performance were clearly seen, including the effect of how small assumptions or errors can progressively accumulate to inaccurately represent the model. With further testing rounds, results can be checked for convergence and modelling errors can be rectified.

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6 Appendix

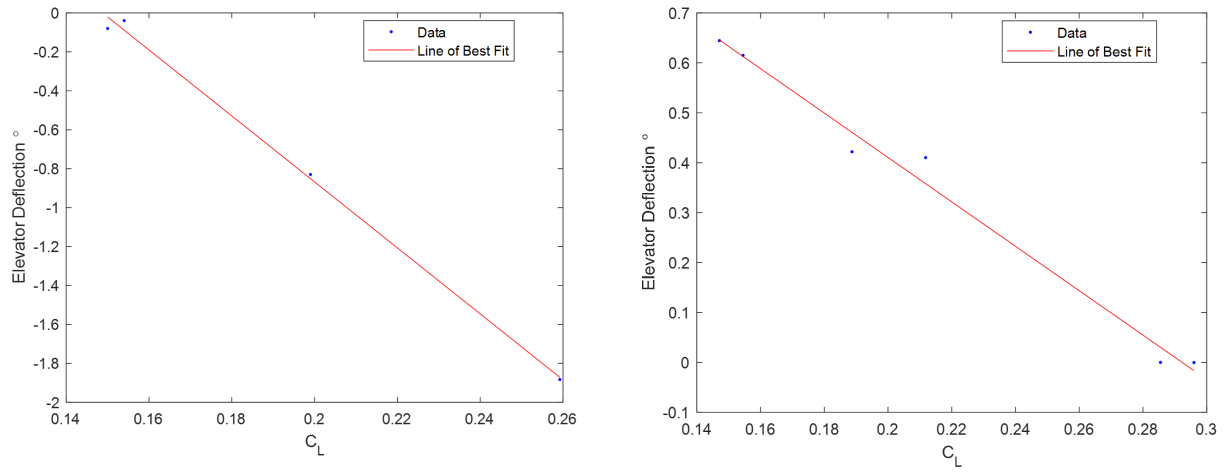


Figure 13: Elevator Deflection vs Lift Coefficient Plots for Static Margin evaluation. (CG positions 0.530 m and 0.600 m, respectively)

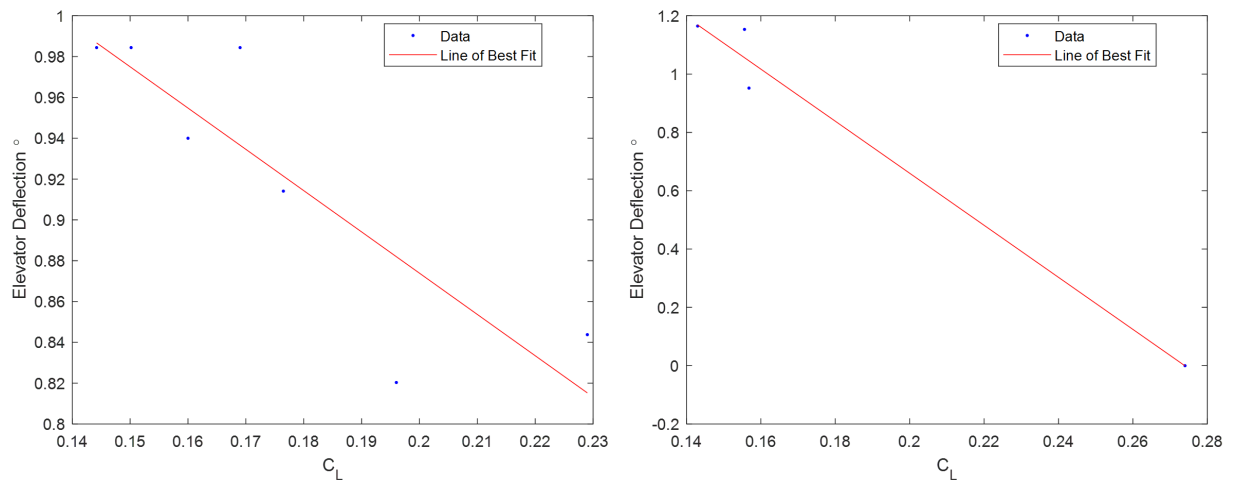


Figure 14: Elevator Deflection vs Lift Coefficient Plots for Static Margin evaluation. (CG positions 0.616 and 0.580, respectively)

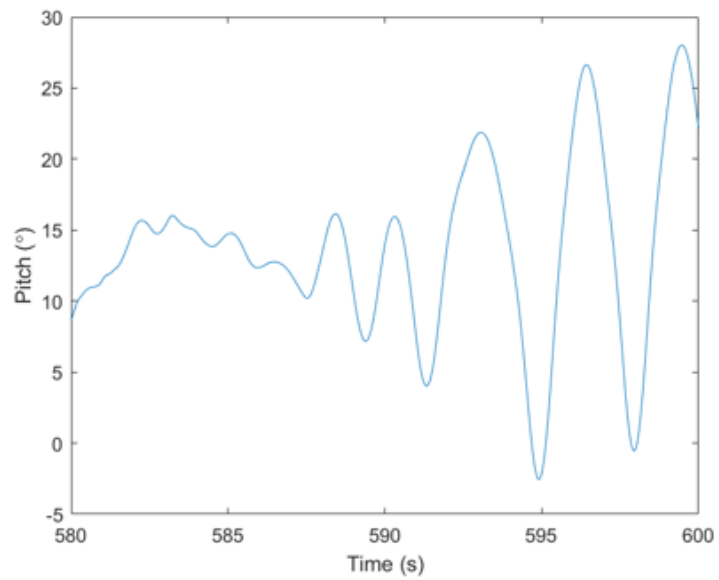


Figure 15: Pitch behaviour at stall observed in simulator.